Assignment - 1

» Section A

1. What is operating system.

Ans: An operating system (OS) is a fundamental software component that manages and facilities the operation of a computer or computing devices.

2. Mention the component of operating system.

Ans: Fundamental component of an operating system are:

1. Kernel
2. User Interface
3. File system
4. Process Management
5. Memory Management
6. Device Drivers
7. File and I/O Management
8. Security and Access Control
9. Networking
10. Utilities
11. Shell and command Interpreters.

3. What is difference between program and process.

Ans: A program is a static set of instruction stored in a storage device while a process is dynamic execution of these instruction in memory with its own resources and stats. Programs became processes when they are loaded into memory and executed. and multiple processes can run concurrently on a computer system.

4. What is Multitasking / Time sharing?

Ans: Multitasking often referred to as time sharing is a computer operating system capability that allows multiple tasks or processes to run concurrently on a single processor or computer system.

5. Differentiate between User View and Kernel View!

Ans: The user view and kernel view represent two different privilage levels within an operating system. User view is where user-level application run with restricted access to system resources. While kernel view is where the operating system core components operate with full access to system resources and control over system operations.

>> Section B

1. What is system call? Explain various system call with examples?

Ans: A system call is a mechanism provided by an operating system that allows user-level processes or programs to request services or functionality from kernel which is the core component of that operating system. Some common types of system calls with example.

1. Process Control :- Process control is the system call that is used to direct the processes. Some process control examples are :- Create process (), Exit process (), Fork (), Exit (), wait, etc.

2. File Management :- File management is a system calls that is used to handle the files. Some examples of file management are : Create file (), Open (), Read (), Read file (), etc.

3. Device Management :- Device Management is a

system call that is used to deal with device. Some example of device management are: Read Console(), write close(), Read(), Write(), Ioctl().

4. Information Maintainence :- Information maintainence is a system call that is used to maintain information. Some example Information maintainence are: Getpid(), Alarm(), set Timer(), sleep(), etc.

5. Communication :- Communication is a system that is used for communication. There are some example of communication are: Create pipe(), MapView of file(), pipe(), Mmap(), shmget(), etc.

6. Protection :- Some examples of Protection are :- Set Permission(), Get Permission(). Allow User Remote devices etc.

2. What is the difference between kernel, Process and multithreads?

Ans:-

Character	Kernel	Process	Multithreading
Definition	The core of an OS	An instance of a running program	A technique that allow a single process to run multitask simultaneously
Managed by	Itself	The operating System	The operating System
Resource	Has its own	Share resource with other processes.	Share resource with the parent process.
Example.	The file system the network stack	A web browser, a word processor.	Multiple thread within a web browser such as tab that load web page

3. List the services provided by operating system in detail?

Ans:- Services provided by operating System are :-

1. Error Detection :- An error is a device may also cause malfunctioning of the entire device. This include hardware and software error such as device failure, memory error, division by zero etc.

2. Accounting :- An operating device collects utilization records for numerous assest and track the overall performance parameter and responsive time to enhance overall performance.

3. Memory Management :- Operating system manages the system memory which is a secure resource. This involves allocating memory to programs keeping track of which memory is in use and remaining unused memory.

4. Networking :- operating system provides networking support which allows computer to communicate with each other over a network. This includes feature such as TCP/IP routing and firewalls.

5. File system Management :- operating system provide a file system for storing and organising files. This includes creating and directing files, managing file permission and providing access to files.

6. Security :- operating system provides security featur

to protect the system and its resources from unauthorized access and use. This includes features such as user authentication, access control and encryption.

>> Section C

1. Explain various types of operating system in detail with example?

Ans: 1. Batch operating System :- This type of OS does not directly interact with the computer. There is an operator which does similar jobs having the same requirements and groups them into batches.

>> Advantage :-
- Multiple user can share batch system.
- The ideal time for batch system is very less.

>> Disadvantage :-
- Batch system are hard to debug.
- It is sometimes costly.

2. Multi programming operating System :- It is simply illustrated as more than one program is present in the main memory and any one of them can be kept in execution.

>> Advantages:
- Multiprogramming increases throughout the system
- It helps in reducing the response time.

>> Disadvantage: There is not any facilities for user

interaction of system resources with system.

3. **Multi processing operating system:-** It is a type of operating system in which more than one CPU is used for execution of Resources. It betters through out the system.

>> **Advantages:-**
- It increases throughout the system.
- As it has several processors, so if one processor fails we can preceed with another.

>> **Disadvantage:-**
- Due to multiple CPU, it can be more complex and somehow difficult.

4. **Multitasking Operating System:-** This OS is simply a multiprogramming OS with having a facility of Round Robin Algorithum. This are of two types:-
1. premptive multitasking
2. Cooperative multitasking

>> **Advantage:**
- It comes with proper memory management.
- Multi program can be executed simultaneously in multitasking OS.

>> **Disadvantage:-** The system gets heated in case of heavy programs multiple times.

5. **Time sharing operating system:-** Each task is given

some time to execute so that all tasks works smoothly.

>> Advantage:
- CPU idle time can be reduced.
- Fewer chances of duplication of software.

>> Disadvantage:-
- Reliability problem
- Data communication problem

6. Distributed operating System:- This type of operating system is a recent advancement in the world of computer technology and are being widely accepted all over the world and that too at a great pace.

>> Advantage:
- Delay in data processing reduces.
- Load on host computers reduces.

>> Disadvantage:-
Failure of main network will disturb the entire or stop the entire communication.

7. Network operating system:- These system run on a server and provide the capability to manage data, users, graphs, security, applications and other networking functions.

>> Advantages: 1. Highly stable centralized servers.

security concerns are handled through servers.

>> Disadvantage:
- Servers are costly.
- Maintainence and updates are required regularly.

8. Real time operating system :- Real time means subjected to real time i.e. respone should be guranteed in specific time. They are of two types Hard Real time OS and soft Real time OS.

>> Advantage:-
- They are less likely to crash or enpirence error as they are reliable.
- They are highly efficient.

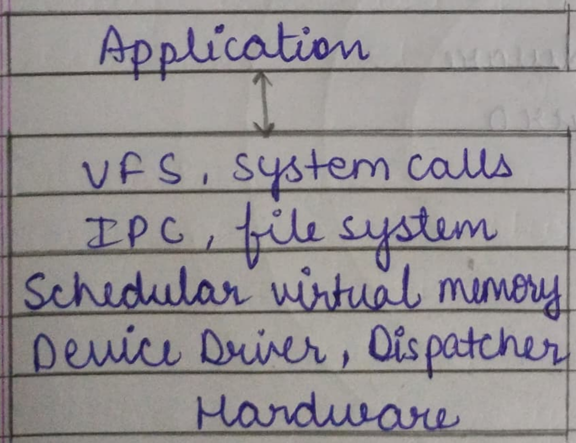
>> Disadvantage:-
- They are more complen than other type of OS.
- They are more enpensive.

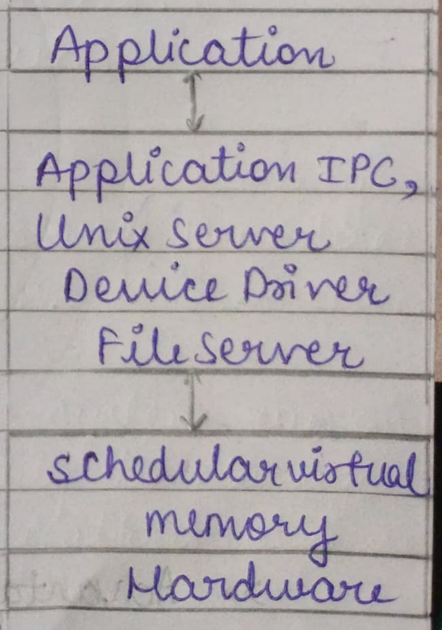
2. Differntiate between Monlithic and Microkernel with Diagram?

Ans:-

	Monolithic	Microkernel
1.	Monolithic kernel is larger than Microkernel.	Microkernel are smaller in sije.
2.	It is not easy to entend Monolithic kernel.	It is easy to entend micro Kernel.
3.	Execution speed is high	Execution speed is low.

4.	In monolithic kernel and kernel services are kept in same address space.	In Microkernel, user service and kernel service are kept in separate address space.
5.	OS is easy to design.	OS is complex to design.
6.	Failure of one component effect or leads to entire system failure.	Failure of one component does not affect the entire system to fail.
7.	Difficult to add new functionalities.	Easier to add new functionalities.
8.	Extra time and resource are needed for maintainence	It is simple to maintain the system.
9.	Example: Microsoft window 95	Ex:- Mac OS X

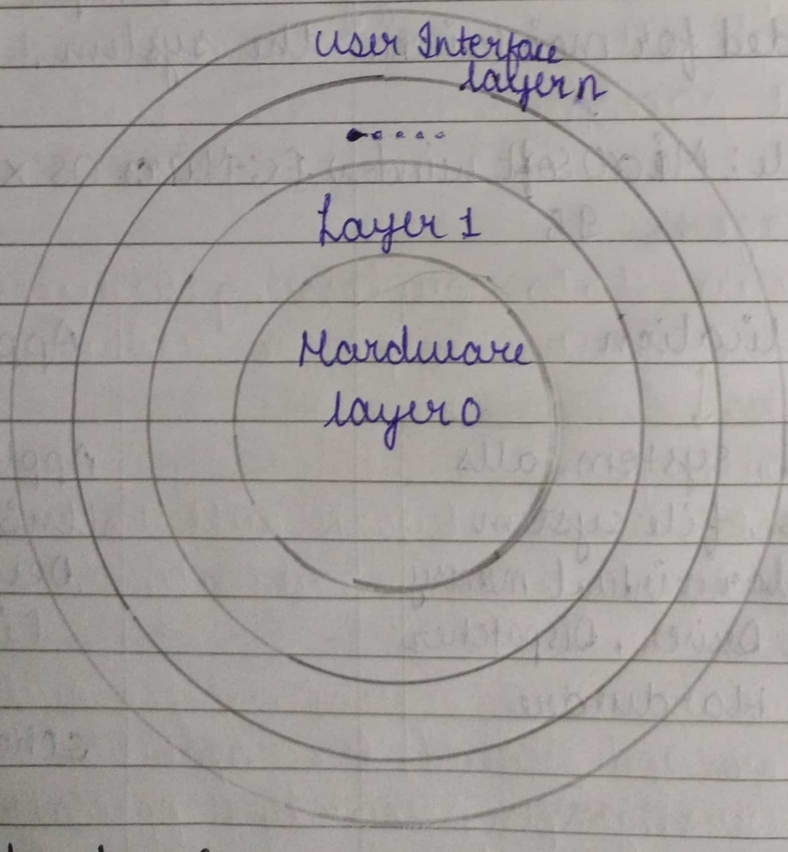




3. Explain layered architecture of operating System.

Ans:- Layered architecture is a type of structure in which the different services of the operating system are split into various layers, where each layer has a specific well defined task to perform. It was created to improve the well pre-existing structure like Monolithic structure (UNIX) and simple structure.

The whole operating system is separated into several layers from 0 to n.
The outermost must be user Interface layer.
The innermost layer must be hardware layer.



>> Advantages:-

1. Modularity :- This design promotes modularity

as each layer performs only the task it is scheduled to perform.

2. Easy Debugging:- As a layers are discrete it is easy to debug.

3. Easy update
4. No direct access to hardware
5. Abstraction

>> <u>Disadvantages</u>:-
1. Complex and careful implementation
2. Slower in execution